

Literature Survey

Predictive Modelling of Chronic Disease Prevalence:

An R-Based Analysis of Regional Health Disparities in India

1.Introduction

The global growth in chronic noncommunicable diseases (NCDs) has resulted in extensive study into predictive modelling methodologies, social aspects, and time-series forecasting techniques. This review pulls together key studies that are directly relevant to examining the prevalence of chronic diseases in India. These investigations include machine learning-based categorisation algorithms, nonlinear autoregressive models, and estimates of epidemic burden.

2.NCD Burden and Epidemiological Background

Using time-series modelling, [3] examined the Global burden of Disease (GBD) 2021 dataset in detail to determine how much the COVID-19 pandemic enhanced the burden of noncommunicable diseases on a global, regional, and country scale. As a result of their research, the number of disability-adjusted life years (DALYs) grew dramatically across South Asia. This supplied the demographic backdrop for the development of the Composite Chronic Disease Prevalence Index (CDPI), which is used in this presentation. The National Family Health Survey Round 5 (NFHS-5) performed by the Government of India in 2024 reinforced these tendencies even further. It revealed significant disparities between states in the incidence of hypertension and diabetes diagnosis, emphasising the importance of state-level longitudinal models.

3.Machine Learning Approaches to Chronic Disease Prediction

A detailed study [8] investigated how machine learning algorithms can be used to detect chronic diseases. It investigated logistic regression, support vector machines, random forests, and deep neural networks on datasets for diabetes, cardiovascular disease, and chronic kidney disease. The authors came to the conclusion that ensemble approaches consistently outperformed individual classifiers. However, they did note that there aren't many longitudinal, multi-state datasets available for population-level surveillance. [9] looked at CKD in particular and compared decision tree and random forest models. They found that

random forests were better at finding early stages of the disease. [19] built on this work by using a multimodal deep learning framework to identify the stage of CKD. This framework combined clinical biomarkers with imaging data to get the most accurate results possible. For chronic liver disease, [12] suggested a mixed machine learning approach that combines gradient boosting with feature selection. This set a standard for modelling the burden of multiple diseases.

4. Time-Series and Nonlinear Modelling

In the time-series domain, [4] showed that Threshold Autoregressive (TAR) models were better than ARIMA at describing nonlinear, regime-switching dynamics in Bitcoin price data. This study directly applied this method to epidemiological series. In the same way, [2] used ARIMA and nonlinear approaches to predict the INR-USD exchange rate and found that two-regime TAR models regularly produce lower RMSE than their linear counterparts on both financial and economic time series. [14] used free and open source R time-series tools to model the economic effects of COVID-19. They set up an open-source process that can be used again and again, which guides the way this case study is done.

5. Socioeconomic Determinants and Policy Context

The role of socioeconomic determinants in chronic disease burden has been extensively documented. NITI Aayog [11] and the Ministry of Statistics and Programme Implementation [10] provide state-level literacy and health index data that reveal a strong negative correlation between educational attainment and NCD prevalence a relationship confirmed empirically in the present study's correlation heatmap. [7] and [13] further demonstrated the clinical utility of ML-based prognostic models for early chronic disease intervention, reinforcing the policy relevance of accurate population-level forecasting tools developed in this case study.

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